

Magic Table - Assignment

Deadline: 23:59 18th October 2025

Introduction

The goal of this assignment is to implement a simple magic trick using a webcam.

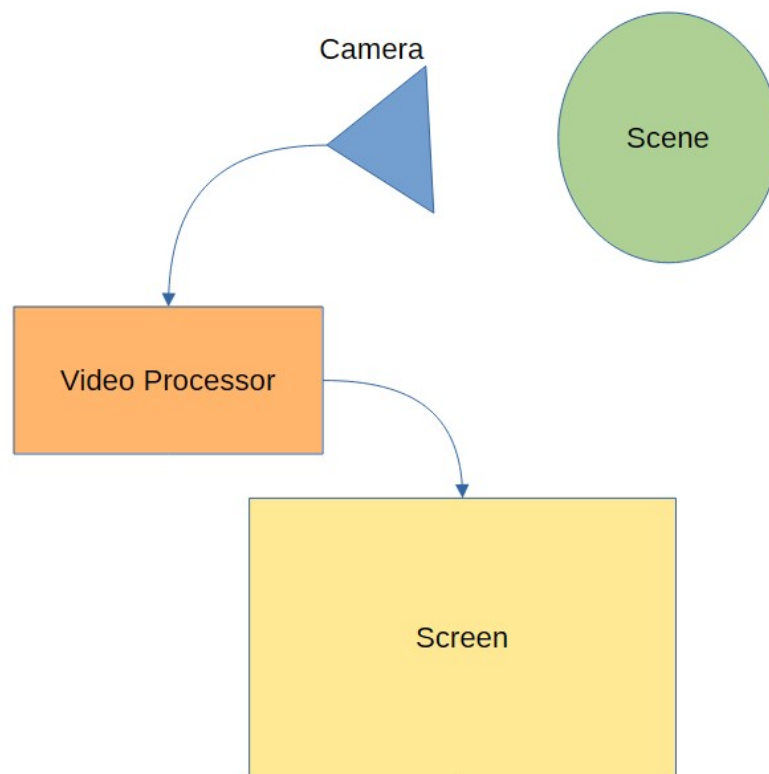
You will demonstrate the control using a small fun marble game.

The Trick

The Magician demonstrates her magical telekinetic abilities by pushing a bottle across a table from the left to the right. There are many ways to create this illusion, but a simple one is that a black tablecloth is used in front of a dark background. The magician uses her legs under the table to ever so gently lift one side of the table, which makes the bottle roll to the other side.



In the assignment, you will implement a high tech version of this illusion. The performance of the trick is virtual. You use the webcam of your laptop to display the screen. Instead of sending the webcam image directly to the screen, you will process it on the laptop and then send it to the receiver.



Vision Processing

To make the trick more convincing, you will have to process the input image so that the angle of the table remains stable, while the bottle is still seen rolling on the table.

There are several methods that can be used to extract the current angle of the table:

- Color calibration of the table edge
- Edge detection to detect the corners of the table on the right or left side
- Feature points (SIFT, SURF, ORB) to detect the rotation of the table
- Template matching for the right and left side
- Difference image between an average frame and the current frame

Discuss the advantages and disadvantages of these approaches and then pick **two** to implement in your assignment.

There are two approaches to solve this problem:

- Reduce the size of the image slightly and then rotate the entire frame so that the angle of the table remains constant
- Extract the table and rotate the table while the magician and the bottle remain unchanged

Implement a version of both methods and the two detection algorithms that you selected above and then compare their performance. You should have a total of four different systems. Which one is more convincing? Which one runs faster.

Submission

To submit your assignment, follow the procedure below.

Honesty Declaration

By submitting an assignment, you claim that your submission is **only your own work**. This means that you have developed the design, implemented the code, run the experiments, and all other work necessary to complete the assignment **by yourself and without the use of any AI tools**.

If you discussed your work with other students, used sources on the Internet, or used AI tools such as ChatGPT, then you **must** say so clearly at the beginning of the assignment.

Use of AI tools is permitted, but you must specify which tools were used and what prompts you entered.

You can discuss parts of the assignment with others, but these discussions must be limited sharing ideas on a whiteboard or notepad. If you exchange code/programs/designs or exchange any material via the Internet, then you must say so in your honesty declaration.

Fill out the [NTNU Honesty Declaration](#) and add any external sources that you used during the assignment.

Submission

The submission consists of three parts: honesty, code, and report.

1. First, create a directory with a name “<Student ID>_assignment_marble”
2. Download and fill out the honesty declaration. Add it to the directory you just created
3. Create a directory “code”, which includes all the source code for your assignment, all build tools, and a file ReadMe.txt which includes all necessary instructions for how to build and run your code
4. Create a directory “report”, which includes a pdf file “<Student_ID>_report_marble” that describes the results of your evaluation of your assignment. How well did it work. What problems did you face when implementing your assignment. Were the results expected or did they surprise you? Did the results highlight shortcomings in the system? Do you have any ideas for how to fix those?

5. Create a compressed archive of your directory, that is a zip file, of your directory. If the file is less than 10 MB, then you can email it to me directly (jacky.baltes@gmail.com), otherwise upload it to a cloud storage provider (e.g., Google drive) and send a link to my email. Make sure that you give read permissions to my email.